Cloud Infrastructure ENG - Take Home Exercise

Objective

The goal of this exercise is to evaluate your ability to design and implement a simple cloud infrastructure using Terraform. You will create a Terraform template to provision the following resources in AWS:

1. **VPC (Virtual Private Cloud)**
   - CIDR block: `10.0.0.0/16`
2. **Subnets**
   - Two public subnets in different availability zones
   - Two private subnets in different availability zones
3. **Security Groups**
   - A security group for the ALB (Application Load Balancer) allowing inbound HTTP (port 80) and HTTPS (port 443) traffic
   - A security group for the EC2 instance allowing inbound traffic only from the ALB security group and SSH (port 22) from the VPC cidr block.
4. **Application Load Balancer (ALB)**
   - An ALB that distributes traffic to the EC2 instance
   - The ALB should terminate SSL (use a self-signed certificate for this exercise)
5. **EC2 Instance**
   - A single EC2 instance in one of the private subnets running a simple web server (use an AMI of your choice)

Instructions

1. **Setup and Configuration**
   - Use the latest version of Terraform.
   - Create a new directory for your Terraform configuration files.
   - Initialize a new Terraform project in this directory.
2. **VPC and Subnets**
   - Define a VPC with the specified CIDR block.
   - Create two public subnets and two private subnets in different availability zones.
3. **Security Groups**
   - Create a security group for the ALB allowing inbound HTTP and HTTPS traffic.
   - Create a security group for the EC2 instance allowing inbound traffic from the ALB security group and SSH from the VPC cidr block.
4. **Application Load Balancer**
   - Define an ALB that terminates SSL. Use a self-signed certificate for this exercise.
   - Configure the ALB to distribute traffic to the EC2 instance.
5. **EC2 Instance**
   - Launch a single EC2 instance in one of the private subnets.
   - Install a simple web server (e.g., Nginx or Apache) on the EC2 instance.
   - Ensure the web server is running and accessible through the ALB.

6. **Output**
    - Configure Terraform to output the ALB DNS name and the private IP of the EC2 instance.

Deliverables

1. A zip file or a link to a public repository (e.g., GitHub) containing:
    - Terraform configuration files (`*.tf` files)
    - A `README.md` file explaining:
        - How to set up and run the Terraform scripts.
        - Any assumptions made during the implementation.
        - How to verify the setup (e.g., accessing the web server through the ALB).